Galaxy workflow for bacterial genomics exploration and chitin utilization enzyme identification: A case study exploring the *Streptomyces albogriseolus* PMB5 genome

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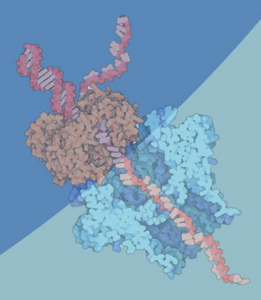

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Abstract

Numerous symbiotic relationships between bacteria have evolved within the insect's gut. Bacterial mutualists within insect guts play a crucial role in modulating the host's metabolism and enhancing digestion to extract maximum energy from consumed food. Therefore the carnivorous insect hosts can be a valuable natural source of chitinolytic bacterial strains. In this study, we have explored and screened for chitin-degrading bacteria within the gut microbiota of the carnivorous insect *Mantis religiosa*. For that reason, we conducted whole-genome sequencing using Nanopore technology of the chitinase-producing *Streptomyces albogriseolus* PMB5 to elucidate the detailed genomic landscape of chitinase repertoire and to explore the genes essential for chitin degradation and their functions using Galaxy platform and a dedicated workflow. Furthermore, our research endeavors contribute significantly to the elucidation of the genomic architecture of *Streptomyces albogriseolus*, a species that has been underrepresented in scientific investigations. This effort not only expands our comprehension of its genetic landscape but also serves for future studies aiming to explore its potential applications in various fields, such as biotechnology, environmental science, and industry. Moreover, our study adds to the Galaxy workflow community by enriching its repository with a dedicated workflow, facilitating easier data analysis for fellow researchers.

Methods

Total DNA was extracted from bacterial strains with QIAamp DNA Microbiome Kit (QIAGEN, Germany). DNA quantity and integrity were checked with a Qubit 4 Fluorometer (Thermo Fisher Scientific, Inc.). Libraries for sequencing were prepared using the Oxford Nanopore Ligation Kit (SQK-RBK114) and loaded onto a R10 flow cell on a MinION Mk1B with a host PC running MinKNOW version 22.10.5 and Guppy 6.3.7. Assembly and annotation of the genome was done using Galaxy EU server https://usegalaxy.eu/

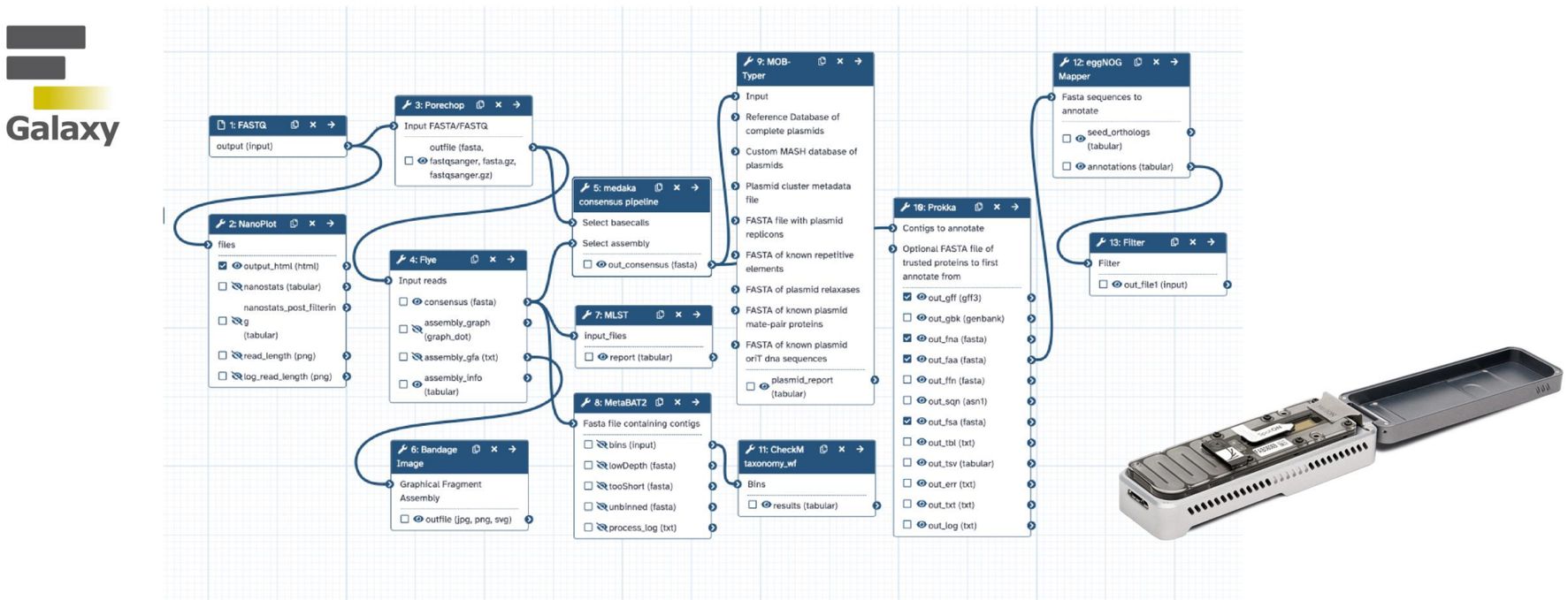

Figure 1. *Galaxy platform workflow for ONT reads assembly, annotation and analysis with Flye, Prokka and EggNOG*

Results

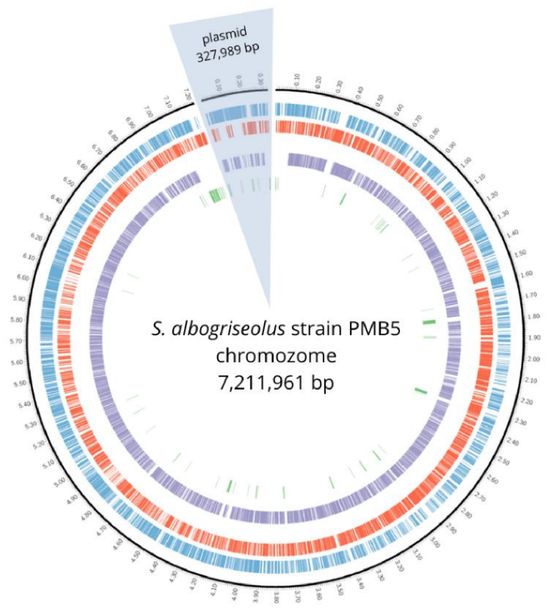

Figure 2. *Map of the S. albogriseolus strain PMB5 genome visualized using Proksee tool - bacterial chromosome (linear) and plasmid (linear). Forward genes are depicted in blue, reverse genes in red, core-genes in purple and strain-specific genes in green.*

CAZy Function class	CAZy Family (No.)
Auxiliary activity	AA0 (1), AA10 (5), AA3 (2), AA4 (1), AA5 (1), AA7 (4)
Carbohydrate-binding module	CBM13 (10), CBM16 (2), CBM2 (10), CBM25 (2), CBM3 (1), CBM32 (3), CBM35 (2), CBM4 (1), CBM48 (4), CBM6 (2), CBM61 (1), CBM67 (2), CBM92 (1)
Carbohydrate esterases	CE1 (2), CE12 (2), CE14 (3), CE19 (1), CE3 (3), CE4 (11), CE7 (2), CE8 (1), CE9 (1)
Glycoside hydrolases	GH0 (2), GH1 (6), GH10 (4), GH11 (3), GH12 (2), GH13 (20), GH146 (1), GH15 (4), GH16 (2), GH17 (3), GH18 (5), GH19 (1), GH2 (5), GH20 (3), GH23 (7), GH25 (3), GH27 (1), GH3 (5), GH30 (1), GH31 (2), GH35 (1), GH36 (1), GH38 (1), GH39 (1), GH4 (2), GH42 (3), GH43 (3), GH46 (3), GH5 (4),GH51 (1),GH52 (1), GH6 (4), GH64 (2), GH65 (3),GH67 (1), GH74 (1), GH75 (1), GH77 (1), GH81 (2), GH84 (1), GH87 (2), GH92 (2), GH95 (1)
Glycosyl transferases	GT0 (1), GT1 (3), GT2 (20), GT20 (1), GT28 (2), GT35 (1), GT39 (1), GT4 (15), GT51 (4), GT76 (2), GT81 (1), GT83 (1), GT83 (1), GT84 (1), GT87 (1), GT9 (2)

Table 1. Table 2 Detected CAZymes and their count in PMB5 genome

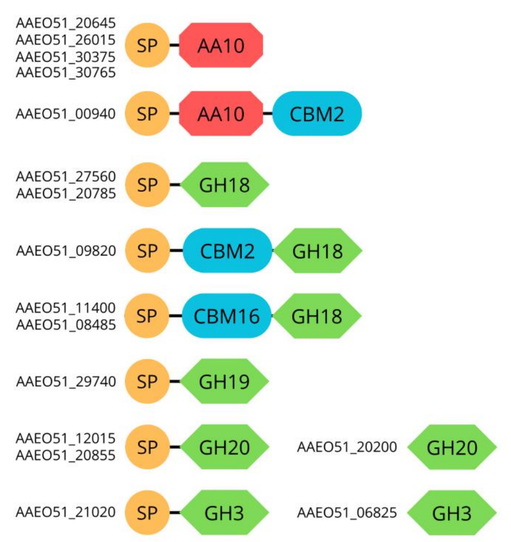

Figure 3. Structural domain organization of predicted CAZymes involved in direct hydrolysis of chitin. SP: signal peptide; GH: glycoside hydrolase; CBM: carbohydrate-binding module; AA: auxiliary activities;

Conclusion

Utilizing the Galaxy platform, we have developed a comprehensive workflow for bacterial genome assembly using long Oxford Nanopore Technologies (ONT) reads and gene annotation, leveraging tools such as Flye, Prokka, EggNOG, among others available on the Galaxy EU server. This workflow enabled the successful assembly of the complete linear genome of *Streptomyces albogriseolus* PMB5 into a single contig, along with an additional linear plasmid. We annotated the bacterial genome using the Prokka tool and subsequently analyzed the protein sequences with the EggNOG tool to extract Glycoside Hydrolases (GHs) and Carbohydrate-Active enZymes (CAZymes) information, crucial for understanding the domain structures of carbohydrate-utilizing proteins. Our analysis identified 56 proteins involved in chitin utilization, including various Chitinases (GH18, GH19), Chitin deacetylase/Acetyl xylan esterases (CE4, polysaccharide deacetylases), β-Galactosidases, β-Glucosidases, Chitosanases (GH46/GH75), Glucokinases, GlcN-fructose-6-phosphate aminotransferases (isomerization), Phosphoglucosamine mutases, and GlcN-6-phosphate deaminase. This study highlights the efficiency of the Galaxy platform for high-quality genome assembly and functional annotation, providing insights into the metabolic pathways and enzymatic capabilities of *S. albogriseolus* PMB5.

Acknowledgments

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